

PlenumHub: Typed Semantic Calculus (Stage 1)

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Abstract

This document formalizes the typed semantic calculus for PlenumHub: spheres, modalities, typed rules, entropy budgets, merge operator and the Media-Quine closure. We provide definitions, a small collection of core theorems, and proof sketches suitable for formalization.

1 Definitions

Modalities and Spheres. Let $\mathcal{K}$ be a finite set of modality keys (e.g., `text`, `audio`, `code`, `proof`). A *type signature* is a finite subset $T \subseteq \mathcal{K}$ declaring required modalities.

A *semantic sphere* is a tuple:

$$\sigma = (I, T, M, E, S)$$

where:

- I is an immutable identifier,
- T is the type signature,
- $M : \mathcal{K} \rightarrow \mathcal{V} \cup \{\emptyset\}$ maps modality keys to values or $\emptyset$,
- $E \in \mathbb{R}_{\geq 0}$ is the semantic entropy,
- S is provenance (a finite DAG of rule applications).

Typed rule. A *rule* r has a signature $r : a \xrightarrow{\epsilon_r} b$ meaning it consumes modality a and produces b while incurring at most ϵ_r entropy increase.

A rule application $\sigma \xrightarrow{r} \sigma'$ must satisfy:

$$M(a) \neq \emptyset, \quad M'(b) = r(M(a)), \quad E(\sigma') \leq E(\sigma) + \epsilon_r.$$

2 Merge operator

Define a partial merge $\oplus$:

$$\sigma_a \oplus \sigma_b = \sigma_m \quad \text{iff} \quad E(\sigma_m) \leq \max(E(\sigma_a), E(\sigma_b)) + \epsilon_{\text{merge}}.$$

If no such σ_m exists the merge is undefined (denoted $\perp$).

3 Core Theorems (informal)

[Merge Idempotence] For any σ , $\sigma \oplus \sigma = \sigma$.

Sketch. Merging identical spheres introduces no new contradictions; choose $\sigma_m = \sigma$ and $\epsilon_{\text{merge}} \geq 0$. $\square$

[Merge Coherence] If $\sigma_a \oplus \sigma_b = \sigma_m$ and $\sigma_b \oplus \sigma_a = \sigma_{m'}$ then $d(\sigma_m, \sigma_{m'}) \leq 2\epsilon_{\text{merge}}$ where d is the metric combining modality divergences and entropy difference.

4 Entropy invariants and Media-Quine closure

Define the Media-Quine operator Q which synthesizes missing modalities using explicitly permitted modality-transducers. Q is idempotent: $Q(Q(\sigma)) = Q(\sigma)$.

Entropy budgets must be paid for synthesis: closing a sphere may increase E by at most the total budget of the transducers used.

5 Next steps for formalization

We recommend porting the definitions to Lean (file provided in ‘src/lean/’) and proving the theorems using a small-library of tactics. The Python reference implementation contains runtime checks mirroring the static invariants.